

Ambient Systems

AI-Driven Compliance & Sustainability Solutions for the Built Environment

Company Overview

Ambient Systems is an AI-driven compliance and sustainability venture re-establishing its Singapore operations to pilot collaborations under BEAMP Cycle 6. Originally spun out of the National University of Singapore in 2021, we've evolved into a global innovation hub with strategic operations across three continents.

- **(HQ)** – Core operations and system architecture
- **Hong Kong** – Asia-Pacific market validation and R&D
- **Québec** – North American research partnerships

Recognition & Patents

WIPO Patent

WO – AI Compliance System for construction verification

Panasonic Winner

Deep Tech Challenge 2023 – Innovation in built environment solutions

Google Cloud

-Up 2022 – AI deployment excellence

Business Activities

We develop AI systems for compliance automation, ESG intelligence, and performance verification across the construction and infrastructure sectors. Our solutions bridge the gap between regulatory requirements, operational excellence, and sustainability reporting.

Construction QA/QC

Automated quality assurance and control systems for building projects, ensuring compliance and performance standards are consistently met.

Infrastructure Digital Twins

Real-time virtual replicas of physical assets enabling predictive maintenance, optimization, and lifecycle management.

Sustainability Analytics

ESG reporting and green finance intelligence tools that track carbon metrics and support environmental compliance.

BEAMP Relevance & Singapore Focus

Built Environment Localization

Adapting our platform for Singapore's regulatory framework and construction standards.

Integrated Innovation

Combining FinTech, GreenTech, and ConTech modules for comprehensive solutions.

Automated Verification

AI-powered ESG reporting and compliance validation at scale.

Global Market Experience

1 **2021 – Singapore/ASEAN**

Early pilots and academic collaborations through NUS

2 **2023 – Japan**

Smart City Conference presentations at City Tech Tokyo

3 **2023 – Hong Kong/China**

R&D expansion and market validation initiatives

4 **2024 – North America**

Research partnerships and pilot deployments

5 **2024 – Europe**

Market expansion and technology refinement

Products & Services

01

BEAMP Cycle 6 POC/Pilot Focus

Our proof-of-concept initiatives target three critical areas where AI can transform construction compliance and sustainability.

- **Concrete Quality** – Automated verification of delivery documentation, slump tests, and cube-test data with real-time alerts
- **Construction Economics** – QA performance metrics linked directly to financial milestones and project cost visibility
- **Sustainability Metrics** – Embodied carbon tracking and materials documentation for ESG reporting requirements

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Platform – ambientsystems.ai

A comprehensive digital ecosystem integrating compliance, analytics, and operational intelligence for the built environment.

- **Financial Metrics Integration** – Connect quality assurance data with cost tracking and budget performance
- **Generative AI Planning** – Scenario modeling for compliance workflows and logistics optimization
- **IoT Connectivity** – Seamless sensor integration with automated data capture and validation

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Consulting – ivandamnjanovic.com

Specialized advisory services leveraging our patented technology and deep domain expertise.

- Civil Engineering performance audits and compliance reviews
- Energy systems optimization and carbon footprint analytics
- Controlled Environment Agriculture (CEA) digital twin development
- Custom IP and system design based on WO patent

Business Targets

BEAMP Cycle 6 represents a strategic inflection point for Ambient Systems—serving as both the anchor for our Singapore-based innovation hub and the launchpad for global deployment across construction, finance, and ESG markets.

Establish Singapore Entity

Launch new legal structure focused on AI compliance and ESG solutions tailored for the built environment sector.

Transfer IP Portfolio

Register complete intellectual property portfolio including WO patent under Singapore operations.

Complete Platform Modules

Finalize development of Ambient Verify and Ambient Twin platform components for commercial deployment.

Secure Pilot Partners

Execute LOIs and MOUs with pilot clients and ecosystem partners across construction and finance sectors.

Advance Technology Readiness

Progress from TRL 6 to TRL 9 through successful BEAMP pilot implementations and real-world validation.

: 2025–2028

2025 Q1-Q2	2025 Q3-Q4	2026-2027	2028
R&D Phase	**Pilot Deployment**	**Commercial Scale**	**Global Deployment**
Platform refinement, IP transfer, pilot recruitment	BEAMP implementations, data collection, validation	Market expansion, partnership development	Regional hubs, enterprise clients

Key Team Members

Ivan Damnjanovic

Founder & Systems Architect

Ivan brings a decade of experience in technical management, risk analysis, and AI deployment, with over five years working across Asia-Pacific markets including Singapore, Nepal, and Hong Kong.

He has led R&D projects under prestigious institutions including NUS, NSERC, and Centennial College, and holds PhD-level expertise in Business Management & Economics.

Patent holder (WO) and lead architect of both Ambient Verify and Ambient Twin platforms.

Languages: Serbian (native), English (fluent), French (intermediate)

David Damnjanovic

Advertising & Communications

David serves as our creative specialist, bringing extensive experience in digital advertising and brand communications through his work at Yourbow.com.

He leads campaign strategy and execution for our online ventures, ensuring our message resonates with stakeholders across construction, finance, and technology sectors.

His expertise in visual storytelling and digital marketing strengthens our market positioning and stakeholder engagement initiatives.

Planned Singapore Hire (2025)

Full-Stack Developer / Data Engineer – We're committed to local talent development and will onboard a Singapore-based engineer to support pilot deployment, platform localization, and ongoing innovation initiatives under BEAMP Cycle 6.

Intelligent Verification System for Concrete Compliance and QA Risk Management

Transforming fragmented construction QA data into real-time compliance intelligence through AI-driven verification and ESG analytics.

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Proposal Summary

Challenge Statement (CS 9): Intelligent Verification System for Concrete Compliance and QA Risk Management

Challenge Statement Owner: Expand Construction Pte. Ltd.

Solution

AI-driven verification platform automating data ingestion, reconciliation, and compliance verification across PDFs, WhatsApp logs, spreadsheets, and e-delivery orders.

Unifies QA/QC, finance, and sustainability datasets into a single dashboard for real-time traceability and ESG-linked performance assurance.

Technology

Unified AI engine combining **OCR, NLP, and machine learning** for automated QA verification with reconciliation logic linking Delivery → Slump → Cube → Approval workflows.

Role-based dashboards delivering instant anomaly detection and ESG metrics from delivery through testing to audit readiness.

Deliverables

- Real-time traceability and audit readiness
- Structured QA datasets for compliance and ESG analytics
- Intelligent risk alerts for claims and approvals

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Solution Overview

The Problem

Concrete QA data is fragmented across WhatsApp, Excel, eDO portals, and emails — making reconciliation slow, error-prone, and unverifiable at audit time.

Expand currently manages **100-200 delivery orders per pour**, each requiring manual cross-checks between DOs, cube tests, and invoices.



Our Solution

Automation

AI/NLP engine extracts key fields from semi-structured documents with reconciliation logic linking DO → Slump → Cube → Approval workflows.

Accuracy

Role-based dashboards for QA, Finance, and Audit teams with configurable alerts for missing or non-compliant records.

ESG Insights

Integration with Autodesk Construction Cloud and supplier portals supporting Singapore's Construction 4.0 and ESG data standards.

70%

Reduction in QA Collation Time

100%

Traceability Across Workflows

Competitive Advantage

Combines engineering domain expertise with explainable AI and patented IP (WO). Leverages **Ambient Verify** (Compliance AI) and **Ambient Twin** (ESG Analytics) modules, designed to integrate with existing tools — not replace them.

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Technology Architecture

Module	Core Function	Technology Stack
AI / NLP Ingestion Engine	Automated parsing of WhatsApp, PDF, Excel, and e-DO data; extracts key fields (DO, slump, cube ID, strength)	OCR + NLP (LayoutLM / GPT) · Python · Pandas
Reconciliation Logic	Links DO → Slump → Cube → Approval; detects mismatches and anomalies	SQL + Rule-Based Engine + ML Consistency Model
Alert & Workflow System	Real-time detection of missing data or non-compliance; triggers notifications	Event Triggers · Threshold Rules · Automation Scripts
Role-Based Dashboards	Interactive visualizations for QA, Finance, and Audit teams; exportable audit reports	React / Streamlit · Supabase / PostgreSQL
ESG Analytics Layer	CO_2 and energy traceability; sustainability and carbon-linked KPIs	Ambient Twin API · Carbon Factor Database · ESG Models
Security & Compliance	Access control, encryption, data audit logs; CSA-aligned data protection	OAuth 2.0 · AES-256 · CSA Cyber Essentials (SG GovTech)

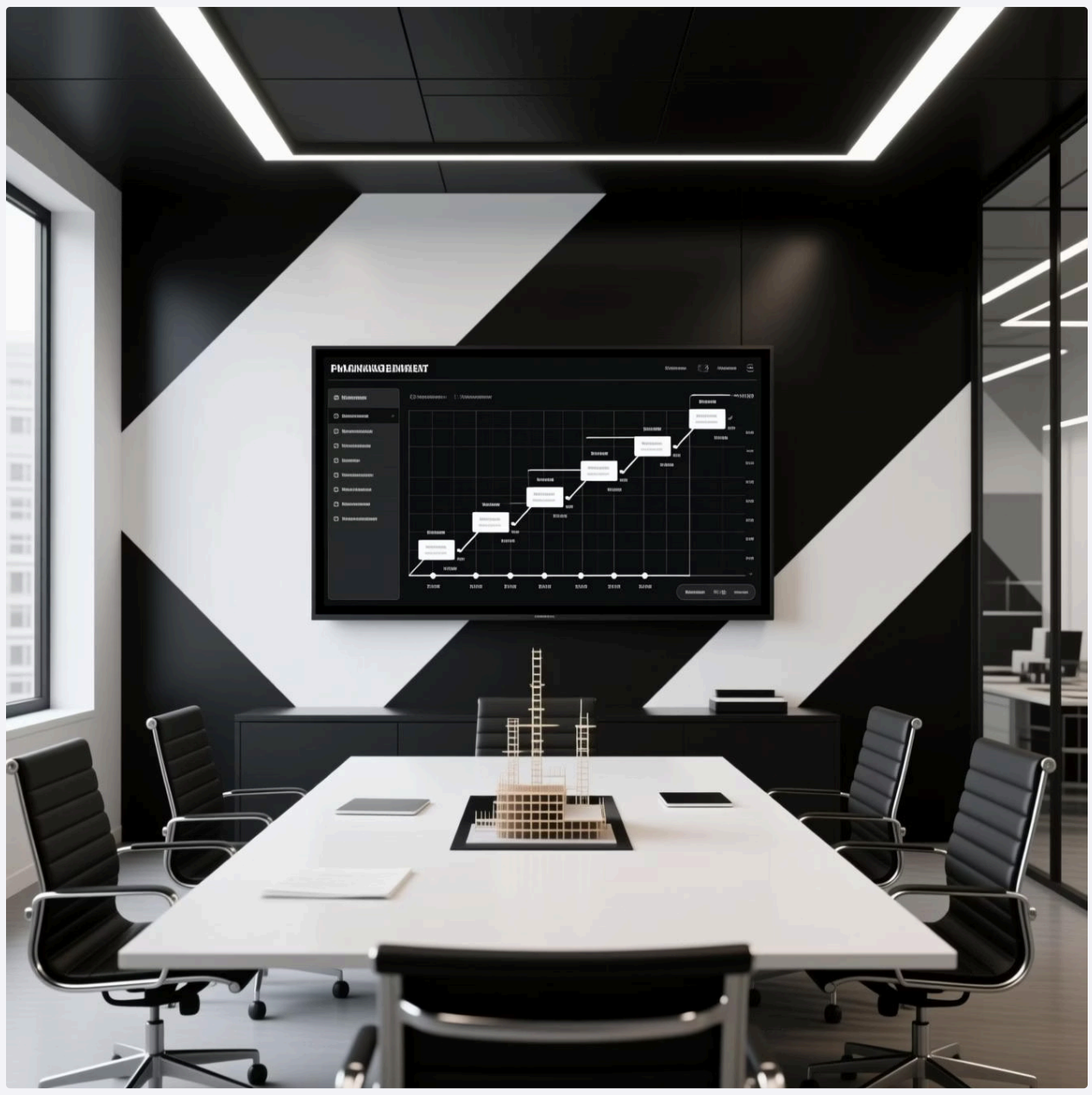
The architecture seamlessly connects data sources through our AI engine to the logic core, culminating in intelligent dashboards with integrated ESG analytics — ensuring security and compliance at every layer.

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POC / Pilot Plan

Objective

Pilot AI-based verification on an active project (NS Square) to validate system performance and ROI.



40-50K

Cost Estimate (SGD)

4

Duration (Months)

Resources

- 2 Data Scientists
- 1 Systems Engineer
- 1 QA Specialist

Deliverables & Milestones

1

Phase 1 (Month 1)

Data identification and organization

2

Phase 2 (Month 2)

Draft ingestion pipeline + collection. Local data formats, writing styles, reporting formats and timings. Scalability considerations for ASEAN environments.

3

Phase 3 (Months 4-5)

Pilot dashboard visualization and live QA traceability interface

4

Phase 4 (Months 6-7)

Validation, , ROI measurement, and reporting

5

Phase 5 (Months 8-9)

data ecosystem for continued use

Stakeholders & Roles

Ambient Systems

AI system design and implementation

Expand Construction

Data access, site validation, feedback loop

BEAMP Ecosystem

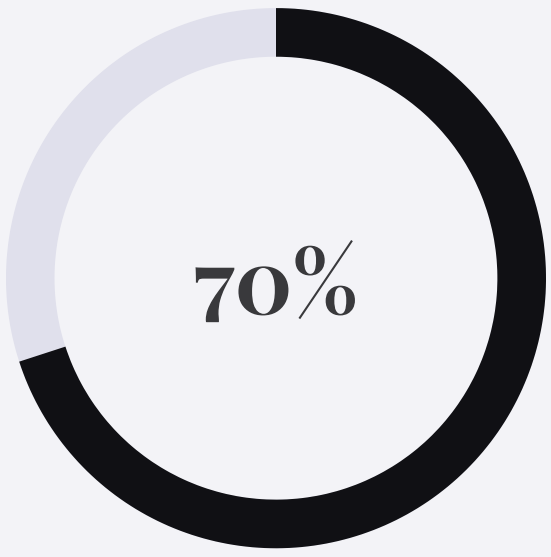
Mentorship and funding support

Test Sites: Online virtually with possibility of using WhatsApp data for real-world validation.

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Impact & Benefits

Quantitative Benefits



Reduction in Manual QA Collation Time

Dramatically streamline data processing workflows



Traceability for Audit and Claims

Complete visibility across all verification stages

Time Savings

Up to **2 working days saved per month** per QA team, allowing staff to focus on higher-value activities and strategic decision-making.

Qualitative Impact

Real-Time ESG Tracking

CO_2 and sustainability monitoring via Ambient Twin layer for comprehensive environmental compliance

Enhanced Confidence

Greater assurance in milestone payments and audit readiness with complete data transparency

Staff Empowerment

AI personal assistants handle repetitive tasks, freeing teams for strategic work

> "From manual verification to intelligent automation — transforming construction QA into a competitive advantage."

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Scalability & Vision

Scope	Plan
Singapore Entity	New Ambient Systems Singapore venture focused on AI compliance tools for built environment
Regional Scaling	Deploy in Hong Kong and Canada for cross-jurisdiction testing and commercial proof
Sector Expansion	Extend to steel, prefab, and BIM compliance modules



Long-Term Vision

Create an interoperable digital twin for compliance — a unified QA and ESG verification layer for Construction 4.0 that sets new industry standards.

Singapore Hub

Establish center of excellence for AI compliance innovation

Regional Expansion

Scale across Asia-Pacific and North American markets

Industry

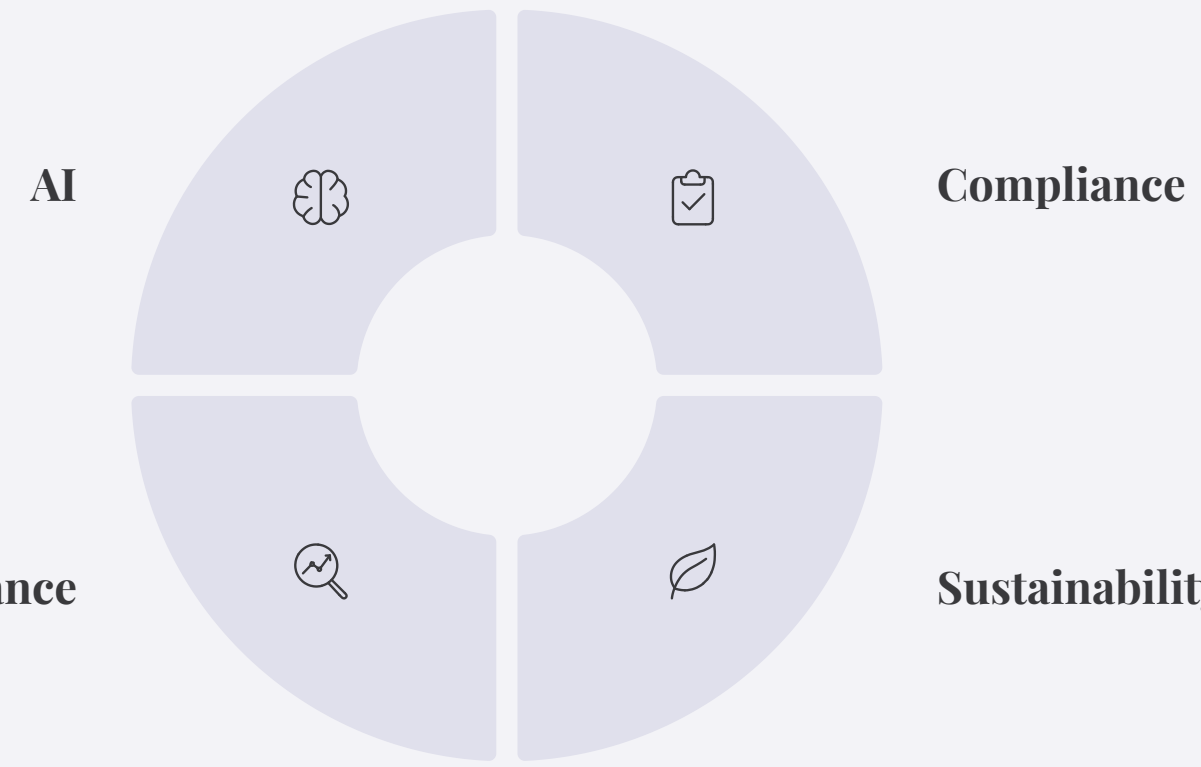
standards for Construction 4.0 compliance

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Turning Fragmented QA Data into Real-Time Compliance Intelligence

Re-establishing **Ambient Systems (Singapore)** to lead the next wave of **AI + ESG transformation** in the built environment.

Empowering contractors, auditors, and regulators with **data-driven verification and sustainability insights.**



Contact

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"From delivery orders to ESG dashboards — making compliance measurable, verifiable, and intelligent."

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